# Whither speech? — the future of telephony

**A V Lewis and F A Westall**

*The last two decades have seen radical improvements in the cost and performance of transmission and switching in telecommunications. To most customers, these changes were only indirectly visible. The next decade will see equally radical developments in things directly tangible to customers — the telephone re-born. After describing historical milestones, this paper debates the future of telephony and reviews the cultural impact of the potential changes.*

## 1. Introduction

### *1.1 A cry from the cradle*

Aaaa-auh-aaa.... mmnn-mmm.... mmnmAH-aaah?

For millions of years, such explorations in babbling conjunction have marked our infant beginning. More significantly, they mark the start of a mental and physical struggle that is long, arduous and mercifully always forgotten — a struggle to master the subtle, highly structured and pivotaly potent ability called speech, an ability that becomes so effortless and integral to our being that we accept its enormous power without a thought. Yet its rich complexity makes us unique among living things, lying at the heart of how we become what we grow to be.

Reflect for a moment how unlikely, even bizarre, the mechanism is. We can, at will, vibrate a tiny piece of our body. We can form, shape and colour these faint vibrations, impressing just a few thousandths of a watt of power on to a thin waft of air. Is it really likely that such insubstantial events could carry joy or sorrow, love or fear, laughter or pain across the vast guilt of consciousness between different minds — and do so in just a few heartbeats? Speech is made in just such a way — and in much less than a minute can penetrate to our deepest emotions of fling our thoughts to the uttermost star. Truly, speech is the cradle of our humanity.

### *1.2 Wither speech?*

Yet some observers claim all human communication will be transformed, within a generation, because of the Internet. Campbell, a Managing Director of Australia Telecom, implies the telephone might be killed [1] by the rapid growth of Internet, data and video traffic. Some say telecommunications will become a low-profit commodity, much like water or electricity supply [2].

The mass media may have just discovered the Internet, but it was conceived 27 years ago [3] and has become a prodigious invention. It can put you in touch with people you need to contact, even though you know nothing about them. It has made a sub-culture out of serendipity (the making of happy, chance discoveries). Negroponte [4] shows how a global computer network can have unique properties, wonderfully augmenting the power and range of human skills. But there are detractors, including some serious and hard-nosed users. Stoll [5] speaks from bitter experience — finding information equated to knowledge, meeting human contact without humanity and being drenched in electronic dross. Unfortunately, the growth and evolution of the Internet is currently so frantic that it addles the brains of a few of those it fascinates. Should we forget how to walk if we learn how to swim?

### *1.3 The power of speech*

The dimensions of human cognition are analysed by Vogt [6] as the key skills of his future breed of ‘knowledge workers’. From Fig 1, it is clear that using only text, data and image-based features, i.e. eliminating the speech-related aspects, would seriously damage or even remove much of the power of human thought. One picture can indeed be worth a thousand words, but the dental pictures that form inside our heads are more potent still — and the most effective expression of those images is through the fast and flexible power of speech.

Speech is more information-rich and revealing than is generally appreciated. Wiseman conducted a test [7] on the relative accuracy of different ways of detecting lying. Observers who just listened to the potential liar speaking did significantly better than those who also watched his picture or those who only read his words. In this large-scale test, the observers with the most clues (those with sound and picture) produced the least effective detection result [1].

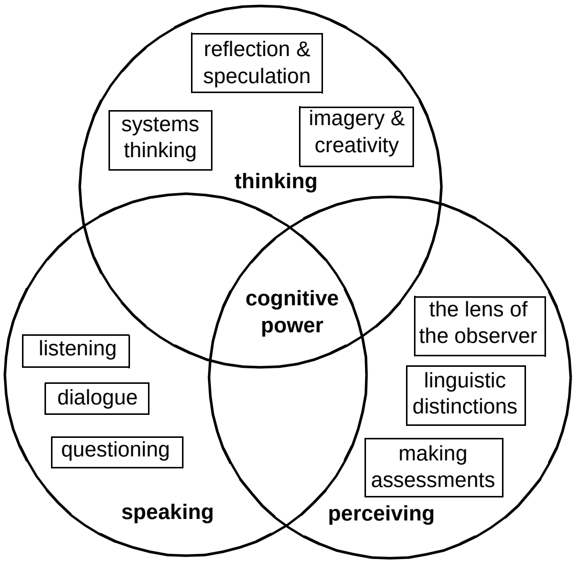


Fig 1 The dimensions of human cognition, from Vogt [6].

The ability to invent metaphor is often considered one of the hallmarks of genius — and linguistic metaphor is, as any good politician knows, the most powerful and memorable of all. The power of speech is something that advocates of new technology for data and image networking, in their enthusiasm, occasionally seem to forget. From the 'Grunters' of the Pleistocene age to the 'Mass Mediators' of today [8], communication by speech has fuelled the ascent of humankind. It is absurd to predict the death of the telephone, but quite plausible to predict its transformation into something that will look and feel quite different.

## 2. The growth of telephony

### *2.1 Birth*

The notion of transmitting sound to a distant place goes back to antiquity. The world's first practical electric device called a telephone (from the Greek, meaning 'far sound') operated digitally. In 1861, Johann Philipp Reis, a Professor in the Garnier Institute of Frankfurt-am-Main, demonstrated an apparatus to his local Physical Society that he called a telephone [9]. It could transmit the pitch of speech sounds, but not their intensity or articulation. It had no true microphone, but used the vibrations of sound to switch an electric current rapidly on and off by a delicately adjusted, diaphragm-driven contact. It was probably David Hughes of Kentucky who proposed the first true microphone, using the variable electrical resistance between iron nails, or carbon sticks, placed loosely in contact. However, Alexander Graham Bell and thomas Augustus Watson (Figs 2 and 3) are universally acknowledged as the true fathers of the 'articulating electric telephone' that we know today.

Fig 2 Alexander Graham Bell.

Fig 3 Thomas Augustus Watson.

[1] Detection by a total of 41 471 television, newspaper and radio observers was 51.8, 64.2 and 73.4% correct respectively (50% by chance) in the 'Megalab Experiment' during BBC Television's 'Tomorrow's World', 25 March 1994.

### 2.2 Childhood

Some key events in telephone history are shown below:

- 10 March 1876 — in a Boston attic, Bell shouts: 'Mr. Watson, come here, I want to see you' — the first intelligible sentence ever transmitted by telephone (Fig 4). Two days later, Bell wrote in his notes: 'The effect was loud, but indistinct and muffled', [10].
- 1881 — the British Post Office grants the first licences for commercial telephony in the UK, limiting service to a five mile radius and extracting a 10% royalty [11].
- 1884 — Clément Ader demonstrates binaural telephony, using two microphones placed some distance apart near the stage of the Grand Opera in Paris. Listeners in the nearby Exposition Hall use two Bell receivers, one to each ear, to experience spatial sound reproduction [9].
- 1891 — an undertaker in Kansas, Almon B Strowger, patents the 2-motion selector for automatic telephone switching [12].
- 1892 — the first automatic telephone exchange opens in La Porte, Indiana — using the Strowger system.
- 1893 — the 'Electrophone Service' sells entertainment by telephone, from London theatres and churches. The service grows modestly until radio broadcasting brings closure in 1926.
- 1912 — the first public automatic telephone exchange in the UK opens at Epsom in Surrey, using the Strowger system.
- 1927 — the first transatlantic telephone calls are made, by radio between Rugby in the UK and Rocky Point in the USA.

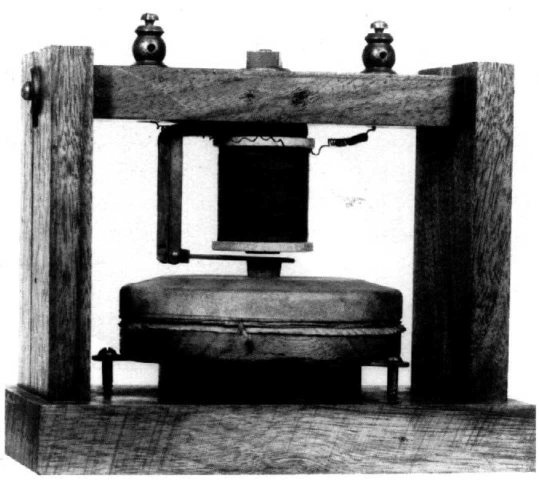

Fig 4 The receiver of Bell's original telephone.

- 1943 — the first amplified undersea telephone cable links Anglesey in Wales with the Isle of Man.
- 1956 — the first transatlantic telephone cable links Clarenville in Newfoundland with Oban in Scotland.
- 10 July 1962 — Lyndon B Johnson makes the first public telephone call via an active satellite in space and says: 'You're coming in nicely...' [13].
- 14 February 1989 — the first satellite telephone call from an aircraft in flight is made, live on BBC Television's 'Tomorrow's World', using the BT Skyphone™ speech coder.

Electronic signal processing floweredearly this century with the work of many people, including Ambrose Fleming, Harold Black, Harry Nyquist, Edwin Armstrong, Norbert Weiner, and John Pierce. Transistors replaced thermionic valves in the 1960s, but the manufacturing tolerances of affordable inductors and capacitors still limited the practical complexity of analogue signal processing in telephony.

The earliest use of digital speech technology in telephony was in the 1960s [14], when pulse code modulation (PCM) systems extended the economic life of junction circuits. PCM coding uses pseudo-logarithmic quantisation[1] of the individual signal samples, for a one-third reduction in bit rate. Digital PCM coding at 64 kbit/s has become the standard for transmission and switching in modern telephony, enabling all the performance and economic benefits of the digital revolution in telecommunications.

In the 1980s, the precision and repeatability offered by digital signal processing (DSP) devices finally removed the limitations of analogue signal processing. This has created a renaissance of signal processing, enabling the economic use of a wide range of complex, adaptive or nonlinear processing algorithms. Such ideas had previously been too cumbersome or too expensive even to contemplate. This rebirth has been breathtaking [16] in its speed and impact on areas as diverse as communications, weapons, entertainment and transport.

1 Both A-law and μ-law codes exploit the same psycho-acoustic effects as the Dolby® B noise reduction system. they are audio or sink coding methods, unlike the more complex and powerful speech or source coding techniques described by Wong et al [15].

### 2.3 The child today

In the past decade, digitisation has enormously improved the economics and flexibility of switching systems. Semiconductor lasers and optical fibres have revolutionised network transmission, but copper is proving resilient in the local loop [17]. Speech coding [15] has helped reduce the cost of international telephony, and is the economic basis of value-added resellers such as Concert ™. Global growth in mobile telephony depends on speech coding, through Skyphone ™ [18] aeronautical or GSM digital mobile systems [19]. Today's scene in speech technology is reviewed by Westall [20] and Johnston [21], while Page [22] and Scahill [23] describe current applications for speech synthesis and recognition technology in telecommunications. Over the same decade, the telephone has been transformed in appearance (Fig 5) and greatly reduced in cost. Modern electroacoustic transducers and integrated circuit technology have been incorporated, but without a fundamental change in specification or performance.

Fig 5 Telephones from Gower-Bell to Slimtel.

### 2.4 Electronic puberty

Just when, as a child, you have worked out how the world operates, when people start treating you like a real individual and when you smile knowingly on hearing the word 'intellectual', along comes puberty and throws the whole pack of cards in the air. The authors believe that puberty is a remarkably good analogy of what the tele-communications industry has begun to experience on an increasingly global scale.

All the ground rules have changed, parts of your body have acquired minds of their own, people treat you the same but expect different results, you treat people differently but expect the same results, anguish is flavour of the month, and you have no clear idea of what happens next; but new and exciting abilities have appeared, with power and potential yet to be fully explored. One such ability is speech technology.

## 3. Telephone evolution

### 3.1 Things to come

The following sections discuss the telephone's success, then examine technical aspects of its future evolution, before reviewing predictions of its cultural impact.

### 3.2 Clairvoyance

'Around the world thought will fly, in the twinkling of an eye,' — Mother Shipton, in the 16th century.

Ordinary mortals only ever see the future through the shifting veil of today. From its birth, telecommunications has seldom foreseen its own future clearly (see Appendix). When some new technology arises it is tempting to think how, in isolation, it could change things. This leads equally easily to wild optimism or blind pessimism. What is difficult to do [24] is to rememberthat everything else can change or interact too. The authors suggest that focusing on the human benefits, rather than the technical capacity, of a new idea provides the best foresight — because human needs change much more slowly than technology.

### 3.3 Variety and adaptability

The telephone is a remarkably successful device. Anyone daring to predict its radical evolution must analyse what customers think is good, and not os good, about the thing. Human needs vary and variety enriches life. Different messages, for different purposes, suit different media. We are remarkably adaptable creatures, moulding our behaviour to the circumstances. We unconsciously adopt quite different psychological attitudes in a hand-written letter, a business fax, an E-mail, with a voice mail message, in a telephone call or during a videoconference.

Only in the videoconference do these attitudes approach the way we communicate face to face, but not every message needs or suits face-to-face contact. Television has become an enormous success. But it has not killed radio, not least because radio listeners are mentally captive while physically free — to get on with something else. Less can be more. Much of the power of conventional telephony, and the appeal of its mobility, lies in the spontaneity and casualness with which calls can be made and received — wherever we are, whatever we are doing and however sloppily or scantily we are dressed.

The authors are sure the evolution of the telephone will be just that — evolution, rather than revolution. New forms and functions will be added, without necessarily abandoning the power and availability of the existing ones. Customers will be most likely to embrace changes which offer greater convenience, improved ease of use or more naturalness in communication. Cochrane [25] describes an ultimate kind of telephony interface that is '... so kind, so beguiling to humans that they can't resist it...'.

### *3.4 Alien though*

So how well does today's telephone perform? Shed all humanity and previous experience. Step into an alien's skin and mind. What might you see and say about telephony?

*Remote communication is their biggest machine, but frustraingly we have not identified the tribe who built it. Its connectivity is impressively rapid and planet-wide, using symbol sequences called telephone numbers. As humans forget these numbers, they often carry a device for linking them to other sequences they do remember, called names. Curiously, their remote communication machine is entirely unresponsive to these names. It is also odd that their remote speech uses a much smaller range of sound than their native mode.*

*When remotely communicating, humans hold an object against one side of their heads. This is shaped like that strange 'bah-nah-naah' fruit, which Om*rty foolishly tasted, although more sombre in colour. This object is common in their communication mythology. The simple task of holding it fully occupies one of their upper extensor protrusions, of which they have only two, for the entire communication. Humans willingly accept this functional loss. Om*try is convinced this behaviour symbolishes an earlier tradition of ritual sacrifice, perhaps in payment for knowledge lying beyond their native senses.*

Handsets are such a familiar and integral part of the telephone that, to some people, the idea of not having one is almost as alien as Om*try. Handsets are clearly preferable for tasks needing privacy or mobility. But not having to hold a handset can be more convenient in some applications and closer to the experience of face-to-face contact. yet loudspeaking telephones, available for over forty years, have had only modest market success — because of acoustic limitations.

### *3.5 The first and last metre*

When optical fibres reach that final domestic doorstep, some communications engineers might feel their dreams have come true — perfect transmission at last. But the real channel in human communication stretches from one mind to another. And sound propagation through the air can be a very nasty channel indeed (Figs 6 and 7), with hundreds of times more fluctuating, multipath propagation than a radio engineer's worst nightmare. This is the technical strength of the handset, placing the electroacoustic transducers so close to the mouth and ear that acoustic transmission is largely free of noise and multipath effects. In future telephony systems, the properties of the acoustic channel will be the dominant technical impairment.

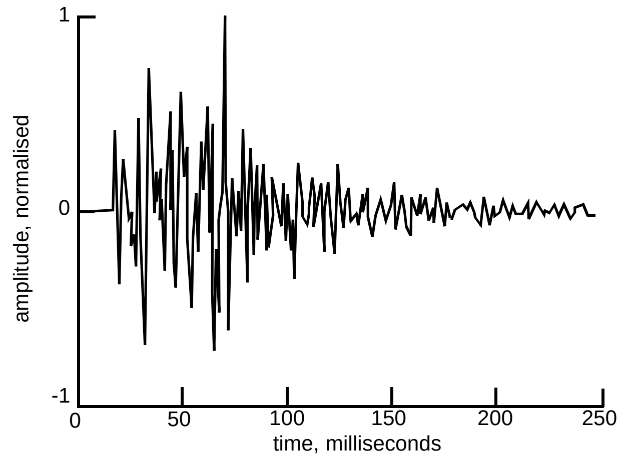


Fig 6 Typical acoustic response between transducers.

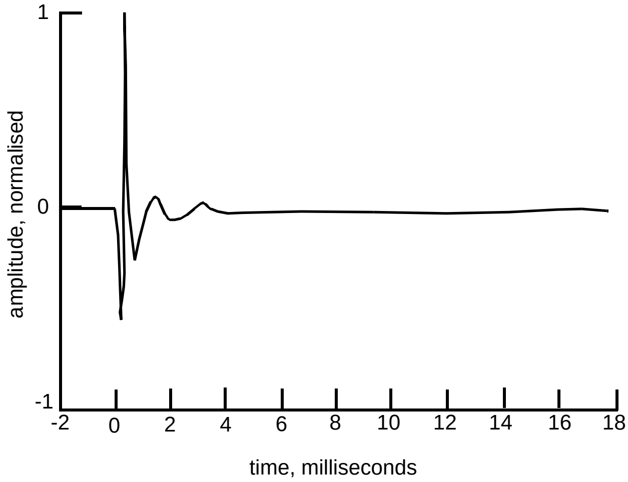


Fig 7 Typical electrical analogue of response in Fig 6.

In loudspeaking or hands-free telephones, for desktop or room conferencing, the transducers are usually better coupled to each other than to the user. This causes two problems. Firstly, a serious risk of loop oscillation and, secondly, multipath contamination of the outgoing speech signal — the 'bathroom effect'. Incoming speech also suffers similarly, but the listener's ear and brain processing essentially removes such contamination.

In 1979, South et al [26] proved the feasibility of solving the first problem with echo-cancelling adaptive filters. This technique [27] provides the same duplex (both-way simultaneous) transmission as a handset and avoids the syllable-clipping degradations of earlier voice-switching methods. The impact of the second problem can be reduced by using a beam-steering microphone array, using an associated set of adaptive filters. This can produce a highly-directional pick-up beam [28], rejecting angularly separate multipath and noise signals, to give a speech quality like that of a handset microphone, but from a distance. Such array microphones could also reduce the impact of environmental noise or nearby conversations on speech recognition systems.

$$f_{max} < \frac{c}{2d} < \frac{(n-1)}{2} f_{min} \qquad \{n \geq 3\} \qquad \text{...... (1)}$$

Equation (1) relates the number of microphones $n$ in a linear array of spacing $d$ with the frequency range $fmax$ to $fmin$ when the speed of sound is $c$. The directionality of a seven-element, experimental array is shown in Fig 8. In practical teleconferencing systems, with wideband hands-free audio, a physically large array of many microphones is implied by equation (1). Such an array could be wall-mounted in a decorative mural, painting or poster. Logarithmic spacing, array weighting or sub-array techniques can be used to avoid an undesirable increase in array directionality with frequency. Efficient algorithms for beam adaption and talker steering in practical environments are currently a subject of international research and development.

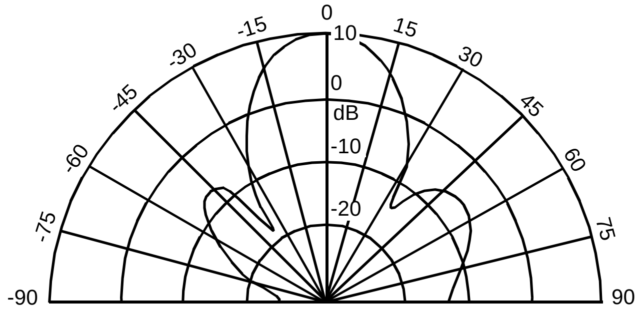


Fig 8 Polar response of an experimental array at 1 kHz.

There are simpler alternatives to holding a handset, such as lightweight cordless headsets, cordless lapel microphones with loudspeakers or cordless earphones with beam-steered microphones. Low-power radio or optical transmission [29] could provide such cordless links.

### 3.6 *Reality limited*

In some situations, a significant delay in replying to a question can imply the opposite of the verbal meaning. As other impairments in speech, data and video communications are reduced or removed, future customers will come to regard excessive delay as infuriating [30] and a significant barrier to natural interactions. This is likely to reinforce the current economic trend toward the use of ground-based links for long-distance and international calls, freeing satellite bandwidth for mobile services.

The quality of conventional telephone sound — fit for purpose, but hardly high fidelity — might seem a major barrier to greater realism, in communication. However, wideband speech coding (with a bandwidth of 50—7000 Hz and a bit rate of 64 kbit/s — G.722) has been technically feasible for many years. It requires new telephones and exchange equipment, but is compatible with existing switching and transmission systems. It can provide much more natural sounding speech, with a useful improvement in intelligibility. The effects of competition and the enormous inertia of the telephone market have prevented its widespread use. The computer and entertainment industries have used wideband audio coding techniques for many years.

### 3.7 *Plural not singular*

The experience of telephony today is rather like buying a piece of fruit — in a world where all the food markets only ever sell any kind of fruit to individual customers. But in future when you buy fruit, the shopkeeper is going to enquire about the precise nature and purpose of the ‘fruit experience’ which you had in mind. What sort of taste, nutrition, texture, vitamin content, colour and size? When, where and with whom are you thinking of eating the fruit? When you reply, the shopkeeper continues: ‘Ah, an apple then. Two of the nineteen varieties of apples we sell meet your requirements. Take your pick.’

The authors believe the future of telephony will be plural not singular. The low cost of new speech and video DSP technology, added to the effects of computer convergence, competition, market liberlisation and high-speed network technology is likely to fragment the current single-specification market. A burgeoning variety of customer equipment and services would then arise, to meet the previously-suppressed variety in the range of human communication needs.

Singular uniformity of specification in telephony has gone unchallenged for so long it seems like deadlock — it was always absurd to consider anything else. This has inhibited improvements in sound quality and limited variety in modes of use. These were tiny sacrifices compared with the mammoth task of achieving efficient global connectivity. Furthermore, uniformity led to one of the most important achievements of telephony today — the universal interoperability of any two telephones anywhere in the world.

A plural set of telephone-with-computer-with-network specifications seems fatal to this vital achievement. Fortunately, the same DSP speech technology that economically enables pluralism can also provide future-proof interoperability, by signal conversion between otherwise incompatible systems. These conversions can add value — for example, a customer using a standard mobile telephone could be a participant in a wideband video conference. Variety need not spell exclusion, merely ‘mixand-match freedom. The future of telephony is not just the telephone.

### *3.8 Face to face and eye to eye*

New ways of using plural mixtures of technology for telephony in the future [31, 32] will need new names, to describe the different but interoperable services that will arise. Such services might include:

- shared-computer-desktop cordless-headset narrowband telephony,
- cross-computer-platform white-boarding with hands-free narrowband telephony,
- wideband hands-free audio and video multi-person-to-person calls,
- high-quality, spatial audio and video conference calls.

These new options arise when multimedia technologies [33], such as shared computer workspaces, eye-to-eye video and immersive conferencing, are integrated with conventional telephony. A remote location could virtually 'appear' through one wall of a conference room (Fig 9). In some circumstances, teleconferencing might become better than physically meeting. Additional hardware or new delegates could readily be added to a meeting, even though their need had not been anticipated — something impractical in a remote physical meeting.

Fig 9 High-quality immersive teleconference.

Developments in speech synthesis and recognition could replace and augment the role of a human operator in simple and well-defined tasks. An electronic 'secretary' could spot key words and recognise important action points in a business teleconference. This 'secretary' would then summarise the items of interest for each participant and prepare a customised text file, annotated with the relevant speech fragments. Speech technology, could also extend the human-computer interface, long dominated by the keyboard, button and screen, by adding an intuitive speech interface. For example, a WEB browser might gain a 'voice-driven, voice-responding' fuzzy search engine, particularly sensitive to the expression of uncertainty. A 'voice helper' might bring conventional display technology closer to the power and flexibility of paper books, by aiding navigation and browsing.

Looking further into the future, imagine using eyes-and-hands-free natural speech, in any language, to instruct intelligent agents about the kind of Internet information you seek. Now imagine roaming seamlessly from reality into a shared, virtual workspace — and being able to see, hear and touch synthetic objects or distant people, as easily and intuitively as in the real world.

All these dreams lead the same way, to a kind of remote communication more like natural face-to-face contact. This is the true destiny of telephony — not in the single mode of narrowband handset speech but in a choice of mode, spanning multiple human senses if appropriate, that addresses previously unsatisfied human needs in communication.

### *3.9 What a piece of work is Man*

The greatest challenge in further improving speech technology lies in gaining a deeper understanding of how human speech works. Today's adolescents might well describe human skills in speech generation, reception and comprehension as 'pretty neat — yeah, wicked'. This phrase is, in itself, a good illustration of the mutative variability of speech, since the message received depends on the receiver. Shakespeare put it more eloquently when Hamlet described humankind as: 'How noble in reason! How infinite in faculty! In form and moving how express and admirable! In action how like an angel! In apprehension how like a god!' (Act II, ii, 317).

To a signal processing engineer, speech is an awkwardly complicated thing [34] and difficult to classify. It is sometimes ergodic but seldom stationary, being at times quasi-periodic and at others quasi-random. The one dependable property of speech is that it varies, not only with different talkers but also with the same talker at different times or in different surroundings. Even the gross properties of speech, such as its loudness, show a significant range of variation.

The way our brains are wired for speech seems to involve multiple layers of complexity. Medical evidence, from patients with disease or accidental brain injury, provides clues about how these functions are arranged in the brain. Grammar, vocabulary, cognition (understanding) and linguistic (speaking) skills seem to be quite distinct and independent. Any one of these functions can be seriously impaired without much impact on the others. For example, some stoke victims suffer 'empty speech', with perfect grammar and linguistic style but severe loss of, say, nouns — despite understanding the 'lost words' when they hear them. Literal and metaphorical understanding also seem to be separate abilities. All speech skills appear to be independent of general intelligence. Children suffering from a condition known as 'Williams Syndrome' show severely retarded intelligence, but often use erudite vocabulary in an exceptionally skilful way.

This evidence suggests that emulating human speech faculties with a machine is an impossible daunting task. But many useful applications of speech technology need only a tiny subset of human skills. Additionally, there is often substantial structure or prior knowledge about a task, which can make implementation feasible.

## 4. Communication culture

### *4.1 Overview*

Recently, many people have forecast the future impact of computer and communications technologies, for different reasons. This section reviews those predictions, debates their impact and then deliberately leaves reality behind.

### *4.2 Convergence*

Some personal computers, especially portable ones, have recently absorbed the modem and its dialler into their hardware and software. Rapid growth in the use of facsimile services and the development of caller identification technology has prompted some manufacturers to add these functions. The next stage in this computer/telephony integration may see the computer swallow the whole telephone. One manufacturer [35] already offers a computer with facsimile, voice mail, caller ID, paging and duplex hands-free telephony.

The computing, entertainment and communications industries are widely forecast [36, 37] to converge, but observers disagree about the ultimate result. All sorts of new services are becoming practical, such as teleconferencing, teleworking, teleshopping and video-on-demand [38—40]. These services raise new and sometimes conflicting issues. For example, guaranteed privacy [41, 42] is needed to prevent commercial or government abuse of personal data without letting criminals exploit secure communications.

The authors believe that the most rapid and dramatic effects of convergence will be seen first in education [43], because of the breadth and depth of information that will become available, which no single family or school could otherwise afford. Children are by nature 'early adapters', with an eager taste for knowledge — who knows where they will lead with the tools we give them?

### *4.3 Cultural change*

The pundits are polarised about the future cultural impact of telecommunications, predicting a world ruled by either Mephistopheles or Gabrriel. The most dire prediction, of the censoring and monitoring of everything with zero personal privacy, makes Orwell's '1984' seem sunny and transforms the Dark Ages into a jolly utopia. The opposing view predicts a triumphant victory of the democratic, libertarian market-place, where the Government disappeared after there was nothing for it to do. Kinney [44] debates both views. Personally, the authors suspect that improving communications will continue to mildly foster democracy and that the politicians have more to fear than the people.

We seem set to enter a 'knowledge worker' world [45], where new technology alters the nature of work itself [6], 'work tribe' alliances blur national allegiances or lead to the death of the nation state [44], and society divides into information-access 'haves' and 'have nots'. The Internet is widely cited as an engine of such cultural change, but with an impersonal or even dehumanised outlook. More natural and versatile telephony, carrying more of the nuances of 'being there', will drive the same cultural changes but with an emphatically human face.

### *4.4 Bit-rate or bountiful?*

Lewis et al [46] have previously discussed two imaginary futures — one was a 'bit-bare' world, with all connections mobile and the transmission of uncompressed signals illegal; the other 'bit-bountiful' scene was the opposite, with mobile telephony illegal and any kind of signal compression laughable. Since that time, the bit-bare future has grown closer, with a marked increase in the use of bit-rate compression for mobile telephony and network economy.

However, there is also evidence of a bit-bountiful trend, with the growth of entertainment services on fixed networks. Negroponte [42] speculates that, in future, everything which now comes over the air will come by wire, while all that now arrives by wire will come over the air. The future may turn out to be both bare and bountiful in bits.

### *4.5 Specific fallacy*

Some say that telecommunications companies will become transporters of bits, without any need or way to add value to the process. Some observers [2] go further and suggest that: 'They are no different than electrical, water, sewer or any other utility competing for the right to push stuff through a pipe'. The authors find this outlook bizarre. Do you tell your water company which specific droplets of water, of all the many in the reservoir, you would like to use next? When poised at a light switch, do you tell the electricity company where you would like the energy to come from? When you make a one minute telephone call, are you happy to receive a 3.75 million of whatever bits first come to hand?

Telecommunications if profoundly unlike water or electricity supply, because of its specificity. Bits are not ‘stuff’ in the sense that water, electricity or sewage is. They are not even ‘stuff’ in the sense of a postal service, since in telecommunications an order of arrival different from that of despatch destroys the message. Bits are not the ‘stuff’ of the product. Meeting human communications needs — in all their rich variety and with planet-wide, technology-independent interoperability — that is the product.

A bit-transport future probably awaits the communications industry when genetic and protein engineering gives an ISO-standard ‘direct-with-brain’ bit port. Until that time, the supply of telecommunications will remain as like the supply of water or electricity as painting a picture is like painting a house.

### *4.6 Fantasy made respectable*

It was with a wry smile that the authors first heard of the work of Peter Schwartz [47]. He has taken the seemingly self-indulgent and turned it into a respectable business practice. Games of wild imagining and idle, speculative fantasy are the basis of his technique called ‘scenario planning’, which has already made profits for Shell UK. Therefore, with almost no sense of guilt, the authors invite the reader to seek truth in unreality and ‘to boldly go’ to the next section.

## 5. Learning from fantasy

### *5.1 Enterprising communications*

When Gene Roddenberry [48] and his team of scriptwriters created the TV series Star Trek (Fig 10) in 1966, they little knew how frequently or how closely their work would be scrutinised. By a combination of imaginative flair and an eye for detail, they transcended the limits of a small budget to produce science fiction that still has something to tell us today. It is not the power of the photon torpedoes nor of the warp drive that astonishes the authors, it is the communications system of the *USS Enterprise*. Its performance so completely realises the potential that today’s speech technology has just begun to reveal.

If Mr Spock saw a 1996-vintage telephone in a time-warp, he would no doubt say: ‘It’s communication, Jim, but not as we know it.’ There are no telephones aboard the Enterprise. There are no handsets, no keypads and no bells — so what is left? In a sense, the Enterprise is one huge telephone. Speech recognition has replaced the keypad and the telephone number — the crew speak to distant people by saying their name.

Fig 10 The crew of the *USS Enterprise* in ‘Star Trek’.
[Picture courtesy of Paramount Pictures Corporation]

In the first *Enterprise*, small boxes on the wall use acoustic echo cancellation to provide two-way, wideband hands-free speech. On the surface of a planet, the crew carry a communicator. This is still not a handset — it is simply opened and held in the hand to both listen and speak. This communicator is more than a mobile telephone. The matter transport system needs an extremely precise three-dimensional location to ‘beam someone up’ and this is greatly assisted if they carry a communication (Fig 11).

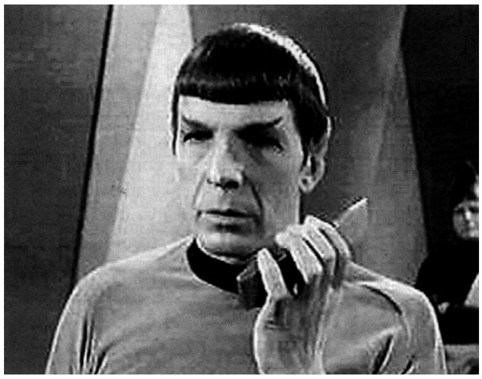

Fig 11 An *Enterprise I* communicator.
[Picture courtesy of Paramount Pictures Corporation]

In the second *Enterprise*, technology has moved on. Multi-microphone, beam-steering arrays are installed on the ceiling of every room, corridor, nook and cranny. These allow unhindered and natural hands-free communication, from any point in the ship to any other.

The speech technology in the *Enterprise II* is provided by hugely distributed massively parallel DSP computers, with a staggering number-crunching capacity. Each microphone array has over 500 tiny diaphragms, with an organic semiconductor DSP device painted on the back of each one. There are a total of 94 248 such arrays, fitted at 6 m intervals throughout the ship, with a combined processing power of approximately 2.4 $\times 10^{15}$ operations per second. The computational requirements of the speech synthesis, recognition and interpretation systems are, of course, very much lower per channel. But these systems have to track many simultaneous conversations, resulting in a broadly similar processing load.

The badges worn by the crew (Fig 12) are not miniature mobile telephones, but merely serve to authenticate the position of their owners to the communication and security systems. If you send a message to a crew member who is asleep, or has said they do not want to be disturbed, the ship automatically stores the message. Unless, of course, the ship's speech recognition and interpretation systems detect a matter of urgency. In that case, the crew member is woken or altered with a chime — the last ghostly echo of the telephone.

Fig 12 The crew location badges in *Enterprise II.*
[Picture courtesy of Paramount Pictures Corporation]

It is interesting to note that despite the ship's ability to hear every word said on board, no person of malign intent has yet breached the security software which protects personal confidentality.

### 5.2 *Comfort and convenience*

Staggering speech technology is used abundantly on the *USS Enterprise*, often to add mere convenience, not facility. Some readers might feel this is a gross abuse of engineering. Will speech technology in future be reduced to the status and functionality of paint? In desktop computers, Microsoft has recently realised what Apple has known since birth — that a machine interface which closely matches natural human behaviour brings benefits that can justify the complexity. Bell knew this too; in 1878 he said: 'The great advantage [the telephone] possesses over every other form of electrical apparatus consists in the fact that is requires no skill to operate the instrument.'

The designers of the *USS Enterprise* show a deep and subtle understanding of this point. They have chosen to put the interests of the crew first, whenever and wherever possible. Speech technology is indeed used abundantly on the *Enterprise*. It makes the lives of the entire crew more comfortable and efficient, by providing a natural and simple match between human and machine skills.

The crew can talk to each other or to the ship's records, wherever they are, with the same effortless ease as to someone standing alongside. Plotting a new course, or targeting an enemy vessel, needs little obvious effort from the crew — because the *Enterprise* can recognise and interpret the relevant words in the speech of the Captain or other key people on the bridge. The ship's computer talks in a voice that not only sounds entirely human, but shows due deference to the Captain's rank — except that, in times of external threat or internal stress, the computer's vocal inflexions, intonation and even choice of phrase change, if necessary, to chide or even rebuke the Captain, as an equal. The synthetic speech system aboard the *Enterprise* clearly embodies aspects of a synthetic persona.

### 5.3 *Optimum interface*

However, the ship's designers also show a sharp appreciation of the occasions when safety or security demands a different approach. If you fire a photon torpedo at someone, you had better mean it. The safe operation of the *Enterprise's* weapons and navigation system demands the utmost level of reliability and integrity, as well as an unambiguous audit trail to show who did what, where and when. That is the reason why the weapons are fired by the antique technology of the push-button and why the helm is not voice-operated but controlled by tactile feedback. Of course, those buttons and levers take your finger-prints and capture a tiny DNA sample, should you operate them.

Sometimes, buttons are best.

### 5.4 *People in charge*

It may have been compelled by a low budget, but all the technology in Star Trek — even the most awe-inspiringly powerful — remains reassuringly human-sized or even invisible. Furthermore, technology is constantly taken for granted in Star Trek and is always secondary to the main theme of interplay between minds — human and non-human. Star Trek machines have to adapt to their users' needs. How different from the viewpoint thirty or forty years before, in films such as Chaplin's 'Modern Times' or Lang's 'Metropolis', where people are subservient, powerless slaves to the great machines.

This inversion of emphasis has already started in telephony. Fifty years ago, customers had to contact the operator for long-distance calls and accept a radical reduction in speech quality. Today, customers can dial calls regardless of distance, with similar expectations of quality. In future, the needs of users will define the machine, rather than the needs of the machine dictating the behaviour of its users. As Cochrane and Westall say [49]:

‘Stop bending people into the technology, and start bending the technology into people’.

Rothenberg [50] uses the phrase ‘deep technology’ to mean a philosophy of deployment that emphasises the interrelatedness of humanity, nature and technology, leading to a potential reformation of cultural attitudes. He stresses how technology, used correctly, can be transparent, extending human insight and enhancing our appreciation of value, in each other and in the natural world. With an outlook that encapsulates the same attitude to technology as Roddenberry’s Star Trek, he says:

‘Technology is empty if it considers only itself. Deep technology ...does not hide us from the world and does not teach us to feel more powerful than we really are.’

## 6. Plotting a course

### 6.1 What’s coming next?

The last section debates intangible issues that may influence or steer future developments in telephony. All the issues concern mental outlook — of customers, competitors, educators and engineers.

### 6.2 Where are we going?

If we stand on the threshold of a dream, how do we get there? The world has changed in all sorts of ways. The energetic commercial competition and multinational business structures. Younger industries, with more flexible and customer-centred behaviour than has been traditional in telecommunications, look hungrily at the potential profits [37]. This situation is quite unlike that which applied during most of the 120-year history of the telephone. A multitude of factors, some new and some not, need fresh examination or re-appraisal in this new world.

### 6.3 Changing expectations

One of the authors can remember his intense frustration, as a child, on opening a plastic ‘Meccano-type’ construction set. There were lots of straight, flat bits suitable for building houses, cars and bridges — but no curved pieces at all, which made it really difficult to build a good chicken. Children can take technology and use it in ways that parents do not dream of.

Equally, telecommunications products are now used in places and in ways that the original designers did not consider. High-speed data modems and facsimile machines are commonplace today, but did not exist when the first national networks were developed. Telephones are now used casually in any room of the house, rather than reverentially in the hall. Public payphones are expected to work well in the noisiest of bars and public places. Yet the electroacoustic characteristics of telephones are largely unchanged from those laid down generations ago.

In an increasingly competitive environment [51], subject to rapid changes in technology, customer expectations may rise or become harder to define precisely before launching a new product or service. But this kind of environment is also one where customers’ views will spell the difference between commercial success and failure. Prudent designers of future telephony systems, despite cost implications, always consider that ‘Meccano chicken’ kind of question.

### 6.4 The mind-set menace

During the multi-faceted and multi-party development of Skyphone™ technology [18], the authors learnt the benefits of a holistic outlook — meaning that the whole can be more than the sum of its parts. This idea sounds more like marketing hyperbole than real engineering, when considered from the ‘linear systems’ viewpoint that is widespread in telecommunications.

When the principle of superposition [52] applies to a system, the whole is always precisely equal to the sum of its parts (Fig 13). The telephone’s history has been dominated by essentially linear and time-invariant systems. That is why it was possible to neatly subdivide telephony into the independently developed domains of switching, transmission and customer equipment. Any one part could be amended or rearranged, independently of the rest, with largely predictable results. This ‘linear’ mind-set has become so widely and deeply ingrained in the industry that it often excludes any other outlook [53].

In the business, as in cooking, no one would dispute that the order in which you do things changes the outcome. Such non-cummutative behaviour is a hallmark of nonlinearity. As explained in section 2.2 new DSP technology has opened a breathtaking panorama of both linear and nonlinear design opportunities. Most of speech technology, such as speech coding, context-dependent speech recognition or sequence-dependent phoneme pronunication, deals with nonlinear systems. New transmission and switching technology, using asynchronous transfer mode or frame-relay techniques, can also show significant nonlinear effects [54].

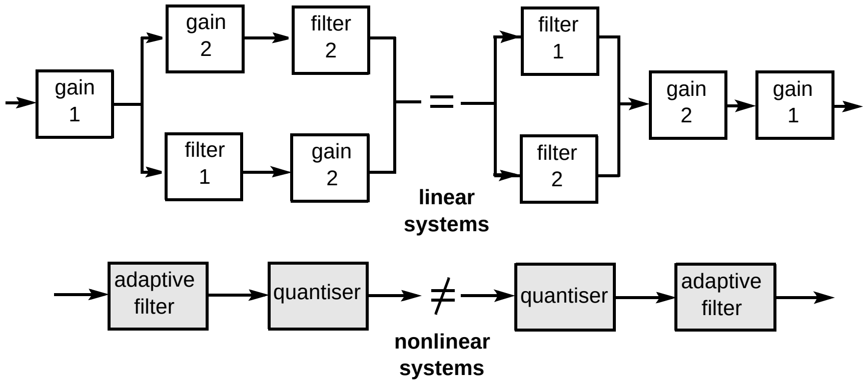


Fig 13 Cascading linear and nonlinear systems.

The preconceptions of the 'linear' mind-set can be a dangerous handicap when dealing with nonlinear technology. If any one part is amended or rearranged, the results may not be predictable. This nonlinear world is reminiscent of the Discworld ® novels of Pratchett [55]. There, life is conventional most of the time except when weird magic breaks out, with unpredictably pleasant or disastrous impact.

There is a significant risk underlying the piecemeal use in telecommunications of systems that are, to varying degree, nonlinear [15, 19]. Those previously independent domains of switching, transmission and terminal design could become uncomfortably interdependent or confusingly less distinct [56, 57]. Delivering predictably good service to customers would then become more complicated.

### 6.5 Hybrid vigour

Someone who studies several disciplines, rather than specialising in one, is sometimes described as a 'Jack of all trades and Master of none', but the disadvantages of over-specialisation were known as early as the 17th century. René Descartes, in his 1629 treatise 'Rules for the Direction of the Mind', wrote: 'If, therefore, anyone wishes to search out the truth of things in serious earnest, he ought not to select one special science, for all the sciences are cojoined with each one and interdependent.' The world of horticulture also knows well the bifour that can result from the cross-fertilisation or hybridisation of living things.

Speech technology is inherently a multi-disciplinary subject, embracing physical acoustics, psycho-acoustics, linguistics, mathematics, signal processing in hardware and software, human factors and computer science. Does the growing importance of such multi-disciplinary technology [58] require a change in the way that some aspects of communications engineering are taught in future? Should it even alter the way in which certain academic disciplines are classified and subdivided?

When an engineer is exposed to a new idea or gadget, the classical reaction is a facsinated 'Ooh, what does it do?' However, those with a background in the arts or humanities have a typical response which is cooler and ultimately more powerful — 'Ah, what can it do for me?' Those two words at the end of that question signal a world of difference, intimately connected with hybrid vigour.

### 6.6 Mindshift

A plural future for telephony has major repercussions on telecommunications planning and design. Instead of solely transporting and routeing bits, networks will need to manipulate them, processing signals to meet customer requirements for interoperability. Pluralism also requires the integration and hybridisation or previously disparate engineering disciplines, in speech, video, human factors, networking, terminal design, information visualisation and more. A radical mindshift in telecommunications engineering will be needed, changing ways of thinking that worked well in the past, but which may work badly in future.

### 6.7 Ways of thinking

'Empires of the future are empires of the mind' — Winston Churchill

Challenges to formerly immutable fundamentals should not really surprise us, because all revolutions inspire chaos. Science is concerned with the pursuit of knowledge, but engineering makes a business of accommodating uncertainty. That is why good engineering is an art as well as a science. If electronic puberty is a good description of our situation, then perhaps we should examine how adolescents cope with the real thing. Is it foolish to suggest that a ‘hang loose’ or ‘let the force be with you’ outlook might sometimes be valuable?

In letting go and embracing subjectivism, perhaps the world of science-fiction can teach hard-bitten, analytic technologists a few useful tricks [59]. When IBM executives first saw the ENIAC computer, they were blind to its potential. Their thoughts were dominated by the physical embodiment — a huge room crammed full of large, heavy, hot and expensive equipment. They though of the manifestation and not the metaphor. they could not separate the function of the device from its form. But such alternate ways of thinking are the foundation of science fiction, which makes logical, rational and self-consistent explorations around imaginative leaps in culture or technology. It is probably no accident that two of the most penetratingly accurate pieces of foresight this century came from the best of science fiction writers — H G Wells about the start of World War II and A C Clarke about geostationary communications satellites.

Telephony has an exciting and challenging future, in which we will have to learn how best to exploit the transforming potential of speech technology — but much more than that, we may need to re-examine, or even re-learn, some of our fundamental ways of thinking [1]. That will be part of both the price and the prize of getting through electronic puberty and becoming adult.

## 7. Conclusions

The childhood of telephony was a time of slow but progressive growth on reliable foundations, with strict parental control by government monopolies. The child was well taught, but adolescence has already arrived — with its rapid, chaotic, miserable and exciting, sometimes frightening, painful and wonderful changes. It is already hard to tell where the former child is going or quite how things will turn out. Strict control is counter-productive [60] and other stronger influences abound — in the effects of fierce competition and peer pressure, accelerating technological change and the convergence of previously distinct markets.

The authors hope to have challenged the reader’s perconceptions and stimulated fresh thinking about the future form and function of the telephone. The original purpose of telephony, ‘far speech’, has become uniquely linked to the current form of the telephone. That form has been so successful that the fundamental metaphor has become obscured. The authors have tried to restore that metaphor to its prime place. Telephony is about the abolition of distance, between one person and another or between people and machines — but, more than that, its true meaning is satisfying the subtle and richly varied needs of human communication.

Predicting the future is usually fraught with failure, but it seems safe to bet that telephones with the appearance of less will accomplish more. Layer upon layer of complex but increasingly affordable speech processing will offer a simple and progressively human-centred interface to a widening range of services.

In Bell’s time, the academic and engineering worlds were ill matched. Bell was not trained in science or engineering, but was driven by a keen and senstitive appreciation of human needs. Today, in telecommunications, dramatic changes await once more. How ironic and mysterious that echoes of these two vital issues, which so fundamentally directed the early days of telephony, seem to have returned.

An information revolution is forecast for the coming century, with communications technology at the forefront of all-embracing change. But this revolution will not just be about better information management. It will reach to the very roots of our humanity, affecting how we individually and collectively learn, think, work and grow.

Adopting both the fervour and the bias of the evangelist, the authors feel confident that speech technology will be the most powerful engine of widespread embrace in this revolution. The authors also believe that meeting and matching the true needs of customers, and of society as a whole, will become a force enormously stronger than any single commercial deal, seductive technology or regulatory restriction. The telephone is on the threshold of multiple transformations — in user-friendliness and convenience, in sound quality and naturalness, and in shape and appearance. These transformations will lead to the largest and most radical shift in the educational, commercial and cultural impact of communications since the telephone was born.

The first commercial talking feature film turned out to be ‘The Jazz Singer’, using manually-synchronised 78 RPM gramophone records to play a few brief songs. Between those songs, Al Jolson ad libbed with some of his stage patter. He chanced to say something remarkably prophetic about the future of cinema sound reproduction. Warner Brothers dared to leave these few snippets of speech in the film they released. Almost seventy years ago, in cinemas all around the world, people sat — some open-mouthed in astonishment, others dewy-eyed with emotion — as they both saw and heard the fiery energy and bubbling enthusiasm with which Jolson shouted [61]: ‘Wait a minute: Wait a minute. You ain’t hear **nothing**’ yet!’

The authors believe that this remark will prove to be even more securely prophetic about the future of telephony. It is indeed the essential message of this paper.

[1] Vogt [6] says: ‘While convergent technologies show significant promise in supporting a radical rethinking of (these) traditional concepts and processes, the effort must begin with our thinking, not with the seductive technologies’.

## 8. Acknowledgements

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## Appendix

*Early vision of telecommunications*

William H Preece was the Post Office Engineer-in-Chief from 1892 to 1899. From a Welsh Non-Conformist background, he was much influenced by Faraday, Thomson and Airy. He gave testimony about the telephone to a House of Commons committee in 1879, saying [11] '...Here we have a superabundance of messengers, errand boys and things of that kind... Few have worked at the telephone much more than I have. I have one in my office, but more for show. If I want to send a message — I use a sounder or employ a boy to take it.'

This quote maligns his vision. He was probably giving his political masters the kind of message he knew they wanted to hear, emphasising short distance operation to deflect criticism. It seems the Treasury wanted the telephone to fail, to defend revenue from the State-owned telegraph service. He had previously been enthusiastic about Bell's telephone. He subsequently co-wrote 'The Telephone' — the first authoritative book on telephone engineering published in the UK. Political interference in telecommunications is now new.

Preece's testimony focused on existing forms of message service: 'Do messenger boys adequately deliver the messages I give them?' His later vital contributions to the growth of UK telephony focused on needs and functions: 'Who might I need to send messages to? Why, when and where?

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Alwyn Lewis received the BA degree in electrical sciences from Cambridge University in 1971, and the MSc degree in opto-electronics from Essex University in 1972. After a short spell working for the Plessey Company on formant-tracking vocoders, he joined BT Laboratories and began work on acoustics, telephonometry and telephone design. In 1978 he helped start a group studying applications of adaptive audio filtering, using hand-wired digital signal processing hardware. He became head of that group in 1984, helping to develop a novel duplex hands-free telephone using a specially designed VLSI chip-set and a programming DSP microprocessor. In 1987 he moved to the adaptive audio systems group, studying a new DSP technique for intrusive measurements of end-to-end telephony performance. He currently leads the speech coding team of the speech technology unit (part of the Centre for Human Communications), where he has been involved with the development and testing of DSP software for Skyphone$^{TM}$ facsimile and data services. His particular interests include speech coding and enhancement, adaptive audio signal processing, echo cancellation, beam-steering microphones and the social history of electro-technology. He is a Chartered Engineer and a Member of the IEE (UK) and (USA).

Fred Westall received a BSc (Eng) degree in electrical and electronic engineering from University College of London and the MSc in communication engineering from the University of Manchester, Institute of Science and Technology in 1973 and 1975 respectively. Following a spell in microwave development, he joined BT Laboratories in 1975 to undertake R&D of speechband modems for the public switched telephone network. He has been closely associated with digital signal processing ever since. In 1982 he became head of speech coding, developing novel speech-and-data multiplexers, which incorporated low bit rate and high-wuality (7 kHz) speech codecs. In 1987 he became manager of data products development where he was responsible for packet terminals and high-speed modem R&D. Recently he has managed the signal processing within the Centre for Human Communications, where he is responsible for downstreaming speechband applications on to DSP speech platforms and for signal processing R&D, in the fields of speech coding, analysis, recognition and synthesis. He is a European Engineer, Fellow of the IEE (UK) and Senior Member of the IEEE (USA).